

IoT-Based Weather Station Using ESP32 and Cloud Dashboard

1. Problem Statement

Design and implement an IoT weather monitoring system that:

- Reads temperature and humidity using a DHT22 sensor
- Sends sensor data to the ThingSpeak cloud platform using ESP32 Wi-Fi
- Displays live graphs of temperature and humidity
- Triggers email alerts when predefined limits are crossed

2. Tasks / Questions to Be Solved

Question 1: System Architecture

Question: Draw and explain the block diagram of the IoT weather station showing ESP32, DHT22, Wi-Fi, and ThingSpeak cloud.

Question 2: Sensor Interfacing

Question: Write firmware logic to interface the DHT22 sensor with ESP32 and acquire temperature and humidity data.

Question 3: Cloud Data Upload

Question: Implement data transmission from ESP32 to ThingSpeak and verify that data fields update correctly.

Question 4: Data Visualization

Question: Configure ThingSpeak to display live graphs for temperature and humidity.

Question 5: Alert Mechanism

Question: Implement an email alert system that triggers when temperature or humidity crosses a set threshold.

3. Tools to Be Used

- **ESP32 Development Board**
- **DHT22 Sensor**
- **Arduino IDE / PlatformIO**

- **ThingSpeak Cloud Platform**

4. Submission Guidelines (Google Drive)

Students must upload the following to **Google Drive** and submit the shared link:

- **Project Report (PDF)**
- **Source Code (.ino / .cpp)**
- **Screenshots of ThingSpeak dashboard and alerts**

Folder Name:

IoT_Weather_Station_<StudentName>